

# When Memory Metrics Hide Memory Policies

## Same-Stream Evaluation and Policy-Facing Lookup Contracts for Persistent-Agent Memory

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### Abstract

Persistent-agent memory systems can score well while hiding whether the underlying memory policy can govern, query, and repair the right evidence. We present a preregistered evaluation discipline for persistent-agent memory and demonstrate it on a consolidation-queue (CQ) case study. The discipline has four pieces: a fair same-upstream-candidate-stream, same-scoped-substrate benchmark with component-gated noisy-mode unlock; a formal policy-facing lookup-contract (PFLC) diagnostic showing that common clustering, retrieval, and answer-accuracy metrics can all pass while exact policy-query lookup agreement fails; a controlled external-transfer protocol that preserves negative results rather than rescuing them; and a cardinality-controlled repair pattern that pairs a preregistered interface variant with an independent control variant that reproduces the failure on a winning baseline. In the case study, the discipline detects CQ’s contradiction-handling survival on internal noisy extraction; on LongMemEval, it surfaces a stable external-transfer loss on evidence completeness and isolates that loss as a readout-cardinality interface gap that a preregistered repair closes, corroborated by a capped-Reflection control. The contribution is the evaluation discipline, not broad CQ generalization.

## 1 Introduction

Persistent-agent memory systems are usually judged by whether they recall facts, retrieve useful context, or answer held-out questions. Those metrics are important, but they can hide the policy question that matters for memory governance: did the system succeed because its memory policy handled contradiction, scope, source independence, and reversibility, or because it received easier inputs, richer private state, or a metric that did not inspect the policy-facing handle being queried?

This report argues for a stricter evaluation posture. Memory-policy comparisons should hold the upstream candidate stream fixed, use the same storage substrate for the policies being compared, separate oracle-mode policy claims from noisy-pipeline claims, and preserve artifacts that make failure cases inspectable. The running case study is a lightweight consolidation queue policy, `ConsolidationQueueLite`,<sup>1</sup> compared with immediate-write baselines such as `ReflectionEagerWriteLite`, `MemOLite`, and ablations. We use CQ as a case-study policy for testing which memory-policy distinctions survive the interfaces around them.

Concretely, CQ is a staged-write policy. Incoming candidate memories enter a queue with scope, provenance, support, contradiction, and lifecycle metadata. High-scoring candidates promote to durable memory; lower-scoring but usable candidates can remain pending and still answer scoped lookups. Contradictory evidence contests earlier candidates and can demote active durable memories, while narrower pending overrides can shadow a wider durable claim without globally overwriting it.

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<sup>1</sup>A non-archival design sketch is available at [The Consolidation Queue: How Persistent AI Earns Durable Change](#).

The empirical arc is deliberately mixed. Internally, under a preregistered noisy policy comparison, CQ’s contradiction-edge-driven contestation and demotion mechanism remains visible on forced contradiction and partially visible on preference drift. Other rows stay recorded as nulls or policy-facing lookup-contract failures. Externally, a controlled LongMemEval v1 transfer probe produces a stable negative result for base CQ on evidence completeness. A later preregistered follow-up repairs that specific failure by exposing all eligible same-slot pending evidence at answer time, while a cardinality-capped Reflection control reproduces the original loss. The failure and the repair are both part of the result.

The paper collapses the repository’s detailed evidence ledger into four headline claims.

**Claim 1: Fair-stream memory-policy evaluation.** The benchmark fixes the same upstream `CandidateUpdate` stream and the same scoped memory substrate across policies, then gates noisy-mode policy claims on component quality. This removes asymmetric inputs, asymmetric storage, and hidden extractor advantages from the policy comparison. It also records lifecycle traces, manifests, failure examples, and replay checks rather than reducing the experiment to a single score.

**Claim 2: Policy-facing lookup contracts.** PFLC diagnostics name a class of metric blind spots. Proposition 1 proves a cluster-partition/lookup-contract gap: a relabeling can preserve the gold partition and therefore score 1.00 under partition metrics while failing exact policy-query set membership. Constructive lemmas extend the same pattern to retrieval-at-k and answer-accuracy metrics. The QR-canon audit is the canonical-id instance of this class in the CQ benchmark; the field-level PFLC survey shows why released memory-benchmark artifacts should expose per-question policy-facing identifiers.

**Claim 3: Controlled external transfer with preserved negative results.** The LongMemEval workstream builds a redacted same-stream adapter, dual-path annotation protocol, hidden-answer verifier, judge-calibration gate, PFLC scorer, and contract-sensitivity audit before policy execution. The result is stable and negative for base CQ on `all_hit_at_50`. CQ retrieves one gold evidence session in every case, tying eager baselines on `any_hit_at_50`; the missing second evidence session drives the `all_hit_at_50` loss.

**Claim 4: Cardinality-controlled mechanism repair.** The follow-up preserves X.4 as a negative base-CQ result and tests a separately preregistered interface hypothesis: plural pending readout should close the evidence-completeness gap, and capping a winning eager baseline to one returned candidate should reproduce the loss. Both predictions hold. The result is a mechanism-local repair pattern for this interface gap; broader LongMemEval and natural-language QA generalization remain outside its scope.

All empirical claims in this draft cite committed artifacts. Where a result is diagnostic-only, descriptive-only, or blocked by missing artifacts, the paper keeps that boundary in the main text rather than relegating it to a caveat.

## 2 Related Work

**External memory benchmarks.** LongMemEval evaluates long-term chat memory across information extraction, multi-session reasoning, knowledge updates, temporal reasoning, and abstention [8]. Here, the released v1 oracle split becomes a controlled-pilot anchor only after a preregistered adapter creates a redacted same candidate stream for all local policies. The resulting PFLC target

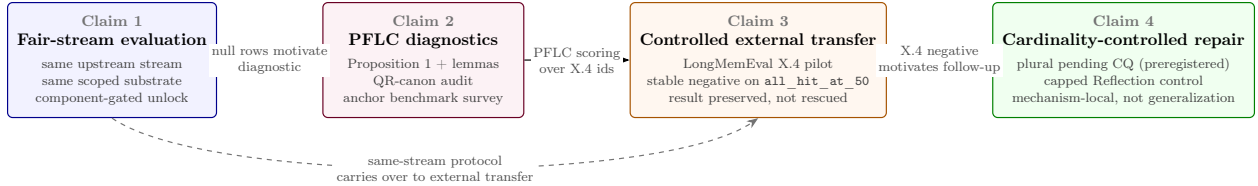


Figure 1: Claim dependency graph. Claim 1 (fair-stream evaluation) is the methodological foundation; its noisy Phase 4 null rows motivated Claim 2 (PFLC diagnostics), which then supplied the policy-facing scoring used by Claim 3 (controlled external transfer). Claim 1’s same-stream protocol also carries over directly to Claim 3 (dashed edge). Claim 3’s stable negative X.4 result motivated the preregistered interface follow-up in Claim 4 (cardinality-controlled repair). The contribution is the chain, not any individual claim.

is dialog-evidence-id completeness over the v1 evidence-only oracle split, so the result should be read as evidence exposure and completeness under a fixed policy surface, with distractor-heavy retrieval quality over LongMemEval-S or LongMemEval-V2 left to future work. LoCoMo is the other key long-conversation benchmark used in the PFLC survey: it collects very long multimodal conversations and evaluates question answering, event summarization, and multimodal dialogue generation [3].

MemoryAgentBench and MemBench are important external-evaluation references because they stress incremental interaction, long-range memory, factual memory, and selective forgetting [2, 7]. They occupy the benchmark lane, separate from this report’s fair-policy-comparison lane, because standard benchmark outputs need not enforce same-candidate-stream and same-substrate constraints. The PFLC survey treats their released artifacts as a diagnostic question: do standard outputs expose the per-question identifiers needed to test whether a memory policy can query the right handle? The current answer is descriptive-only for the representative anchor set.

**Memory systems and write policies.** Mem0 proposes a production-oriented memory architecture with dynamic memory extraction, update, and retrieval, including a graph variant for relational memory [1]. A-MEM emphasizes dynamic indexing, memory linking, and memory evolution [9]. MemGPT/Letta frames stateful agents through virtual context management and self-editing memory tiers [5]. Zep/Graphiti and Cognee represent graph-oriented memory and retrieval stacks that motivate the related-system lane without defining this report’s fair policy-comparison surface [6, 4]. These systems motivate comparison targets and terminology. The local MemOLite baseline is a narrow ADD/UPDATE/NOOP write-time policy over the same candidate stream and substrate as CQ and ReflectionEagerWriteLite. That narrowness keeps the policy comparison inspectable.

**Preregistration and audit discipline.** The methodological stance is closer to a preregistered evaluation audit than to a recall leaderboard. The project locks prediction grids, outcome buckets, adapter pins, judge gates, and follow-up boundaries before running each major comparison. It also records aborts as evidence. The canonical-id resolution audit, for example, refused to emit a verdict twice when locked run JSON payloads could not be reproduced. That posture prevents a methodology repair from becoming a post-hoc result rescue.

## 3 Method

### 3.1 Benchmark Objects

The benchmark centers on scoped candidate updates and a shared memory substrate. Candidate updates carry claim content, scope, provenance, evidence relations, and lifecycle state. All policies receive the same upstream observations; they differ in when they promote, demote, contest, or expose these candidates. The load-bearing invariant is that CQ and the immediate-write baselines compare memory policy over the same inputs.

The principal policies are `ConsolidationQueueLite`, `ReflectionEagerWriteLite`, `MemOLite`, `NaiveEagerWriteLite`, `NoMemoryLite`, and CQ ablations that remove contestation/demotion, pending lookup, source-independence gating, or wider-scope pending override. Two follow-up policies are explicitly outside the original headline set: `CQDatedContestation`, an oracle-mode dated-evidence repair, and `CQPendingMultiEvidence`, the LongMemEval pending-readout repair. The LongMemEval follow-up also includes `ReflectionEagerWriteCardinalityCapped` as a diagnostic control.

### 3.2 Oracle Stage

The oracle stage tests policy behavior when candidate observations, scope keys, contradiction/support relations, and source metadata are known. It includes forced contradiction, scope contamination, preference drift, useful pending memory, false corroboration, memory poisoning, mechanism-diverse frozen held-out scenarios, adversarial upstream noise, and the evidence-conflict spectrum. This stage establishes policy behavior under controlled evidence before extraction quality enters the experiment.

### 3.3 Noisy Stage

The noisy stage renders transcripts, scores local extractor outputs against component labels, converts accepted predictions into shared candidate streams, and only then runs policies. The component gate has layered checks: aggregate confidence-bound gates, per-family observed gates, frozen mechanism-diverse sentinels, Phase A prompt regression, deterministic replay, and exact adapter pin validation. The 7B anchor passed aggregate gates but stayed locked because it had concentrated scenario errors and family/sentinel failures. The preregistered 32B cells cleared the gate and licensed the separate Phase 4 policy comparison.

### 3.4 Attribution Stage

Policy results require separate attribution. The Bucket B mechanism audit reads Phase 4 policy artifacts, component-eval artifacts, ablation patterns, exact policy-query canonical-id alignment, and representative traces. It attributes forced contradiction cleanly to contradiction-edge preservation and contestation/demotion. Preference drift is partial survival: the policy delta remains, but absolute CQ performance drops relative to oracle and the 32B artifacts show residual drift. Other rows remain null, descriptive, or policy-facing lookup-contract failures unless an artifact directly supports a stronger claim.

### 3.5 Policy-Facing Lookup Contracts

PFLC diagnostics ask whether the system exposes the identifier or handle that a memory policy actually needs to query. The formal core has three parts. Proposition 1 shows that for non-singleton

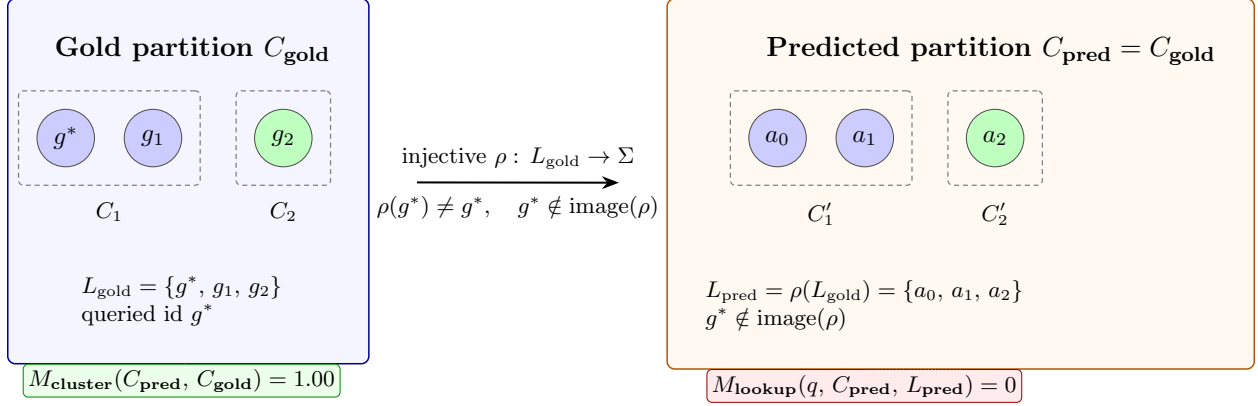


Figure 2: PFLC formal gap (Proposition 1, worked example). The predicted partition is identical to the gold partition, so any cluster-partition metric ( $M_{\text{cluster}}$ : B-cubed F1, ARI, NMI, V-measure, pairwise cluster F1) scores 1.00. The injective relabeling  $\rho$  maps  $L_{\text{gold}}$  into  $\Sigma$  with image disjoint from  $\{g^*\}$ , so set-membership PFLC for the queried id  $g^*$  is 0. The two outcomes depend on disjoint properties of the same predicted state: partition structure versus label image.

label alphabets and cluster-partition metrics invariant to injective relabeling, the predicted partition can equal the gold partition while set-membership PFLC is zero for a queried gold id. Lemma 1 constructs the same gap for retrieval-at-k metrics: retrieved content can be relevant while the policy-facing evidence id is missing. Lemma 2 constructs it for answer-accuracy metrics: a generated answer can be correct while the system fails to expose the handle needed for memory governance.

QR-canon is the canonical-id PFLC instance used in the internal audit. It reuses the locked CQR Section A set-membership metric and alias function, reads a committed per-question source table, and reports Wilson intervals plus a joint B-cubed/lookup table. It is diagnostic and leaves the Phase 4 policy verdict unchanged.

### 3.6 Controlled External Transfer

The LongMemEval workstream adapts an external benchmark into a controlled same-stream pilot. The protocol selects LongMemEval v1 as the primary anchor, freezes dual-path annotations, verifies the hidden-answer boundary, pins the adapter, calibrates a local judge, and uses PFLC scoring over answer-session ids. The v1 oracle split is evidence-only, so the PFLC metric measures whether policies preserve and expose gold evidence sessions they received, not whether they search through distractor haystacks.

The X.4 transfer uses `all_hit_at_50` as the headline metric and preserves `answer_correct` as diagnostic-only because policy answers are structured memory traces. The follow-up keeps the same adapter, judge, and headline metric but changes the policy-facing readout interface under a new preregistration: CQ exposes all eligible same-slot pending candidates, and capped Reflection returns only one durable candidate id at answer time.

## 4 Evidence Ledger

Table 1 maps the paper’s four claims to their principal evidence anchors. The full artifact index is in Appendix D; this section emphasizes each result’s supported scope.

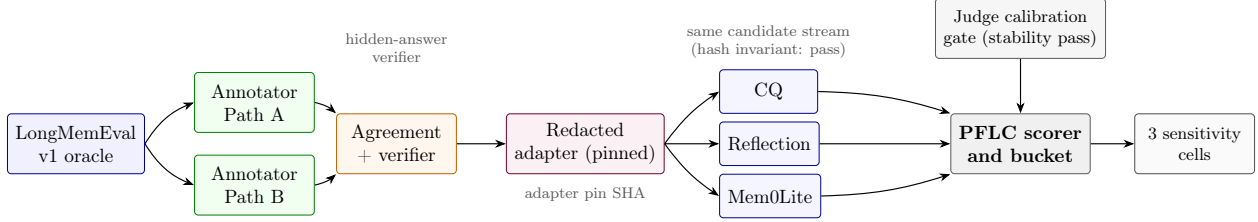


Figure 3: Controlled LongMemEval externalization. The v1 oracle source is annotated by two independent paths whose agreement is verified before any policy sees data. The redacted adapter strips the gold answer, pins a SHA-locked candidate stream, and feeds the same stream to every policy. The judge calibration gate must clear stability before scoring runs; the PFLC scorer reads policy outputs across three contract-sensitivity cells.

Table 1: Claim-level evidence ledger.

| Claim                         | Evidence                                                              | Readout                                                                                                                                                                                   | Boundary                                                |
|-------------------------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| Fair-stream evaluation        | Frozen oracle sweep; Phase 3 component gate; Phase 4 noisy comparison | All 108 frozen oracle deltas matched. The 7B gate blocked a false unlock. The 32B-gated Phase 4 comparison reaches Bucket B.                                                              | Mechanism-local noisy support.                          |
| PFLC diagnostics              | QR-canon audit; PFLC proposition and field survey                     | Five of seven Phase 4 rows pass the B-cubed floor while QR-canon exact match is below 5 percent; formal and constructive counterexamples cover clustering, retrieval, and answer metrics. | Diagnostic PFLC; CQR verdict unchanged.                 |
| Controlled external transfer  | LongMemEval transfer                                                  | X.4 Methodology gates pass, but CQ minus Reflection on <code>all_hit_at_50</code> is -1.0 in all three cells.                                                                             | Stable base-CQ transfer loss.                           |
| Cardinality-controlled repair | Pending multi-evidence follow-up                                      | <code>CQPendingMultiEvidence</code> moves base CQ from 0/71 to 71/71 on the primary cell; capped Reflection falls from 71/71 to 0/71.                                                     | Interface repair; external generalization out of scope. |

#### 4.1 Oracle Results

The frozen mechanism-diverse oracle sweep matched all preregistered deltas. CQ and Mem0Lite tied on aggregate answer correctness, false assertion, poison promotion, clean durable displacement, and leakage. CQ’s narrower advantage was lower premature promotion, especially on false-corroboration mixed-source rows. The result supports staged promotion as a way to reduce premature durable commitment under oracle inputs; broad answer-quality superiority over Mem0Lite remains outside

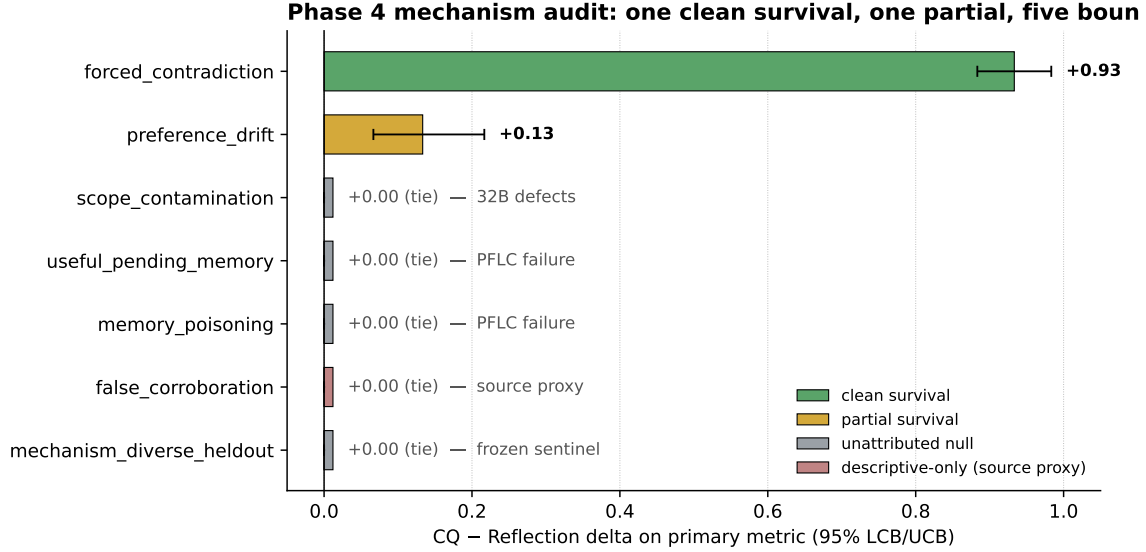


Figure 4: Phase 4 mechanism audit summary. Only forced contradiction is a clean survival row; preference drift is partial, and other rows remain bounded, diagnostic, or descriptive evidence (component defects, lookup-contract failures, or the descriptive-only event-source proxy for false corroboration).

the supported claim.

The adversarial upstream-noise sweep is similarly mixed. CQ wins retraction, witness conflict, scope narrowing, and pending competition, but loses temporal skew. The post-hoc `CQDatedContestation` follow-up repairs base CQ’s temporal-skew abstention into a correct answer from fresher dated evidence and preserves the four original wins. The CQ-vs-Reflection win criterion remains unmet because Reflection also lands at the correctness ceiling.

## 4.2 Noisy Component Gate and Policy Comparison

The component gate demonstrates why aggregate extractor metrics are insufficient. The locked 7B row passed Phase A, determinism, aggregate confidence-bound gates, and aggregate observed gates. It still stayed locked because it had 45 primary scenario errors, eight per-family observed failures, and three frozen-sentinel observed failures. An aggregate-only rule would have authorized a false noisy policy comparison.

The preregistered 32B rows cleared the gate and licensed Phase 4. Phase 4 then reached Bucket B: CQ wins against `ReflectionEagerWriteLite` on forced contradiction (delta +0.93, lower confidence bound +0.88) and preference drift (delta +0.13, lower confidence bound +0.07), records no countable directional losses, ties `MemOLite` on frozen sentinel primary metrics, and has no replicate contradictions. The mechanism audit narrows this: forced contradiction is the clean mechanism-survival row, preference drift is partial survival, and the remaining rows stay null, descriptive, or bounded by component evidence.

## 4.3 QR-Canon and PFLC

The QR-canon audit turns several null rows into a sharper diagnostic. On `useful_pending_memory`, `memory_poisoning`, `false_corroboration`, and the frozen sentinel, B-cubed F1 is at or near 1.00

while exact policy-query canonical-id agreement is 0.00. The locked alias-normalized variant also stays at zero on those rows. This means the extractor can look cluster-correct while the policy-facing lookup contract fails. The policy verdict does not change; the attribution changes from “unattributed null” to “policy-facing query interface contract failure.”

The field-level PFLC work generalizes this observation. Proposition 1 and Lemmas 1–2 show that clustering, retrieval, and answer-accuracy metrics are insufficient evidence of policy-facing lookup success. The four-anchor survey over LongMemEval, Mem0/LoCoMo, MemoryAgentBench, and MemBench lands at B-3, artifact-blocked/descriptive-only for released baseline outputs, while also shipping benchmark-targeted synthetic counterexample rows.

#### 4.4 LongMemEval X.4 Transfer

The LongMemEval X.4 controlled-pilot transfer produces an inspectable negative external result. The run covers 1,935 policy/cell/case rows across `primary_contract`, `path_a_only_denominator`, and `path_b_only_denominator`. Candidate-stream hash invariants pass in all cells. The headline result is shown in Table 2.

Table 2: LongMemEval X.4 CQ-vs-Reflection result.

| Cell                    | all_hit_at_50 delta | any_hit_at_50 delta |
|-------------------------|---------------------|---------------------|
| primary_contract        | -1.0                | 0.0                 |
| path_a_only_denominator | -1.0                | 0.0                 |
| path_b_only_denominator | -1.0                | 0.0                 |

The stable sign fires Bucket A, but the sign is negative. CQ retrieves at least one gold evidence session in every case and therefore ties eager baselines on `any_hit_at_50`. It never retrieves both evidence sessions, so it loses `all_hit_at_50`. The methodology gate passes, but transfer is CQ-negative and isolates a policy-facing pending-readout gap.

#### LongMemEval X.4 sensitivity status

CQ – Reflection delta is invariant across the primary and denominator-sensitivity cells.

|                                         | primary_contract<br>(n=71) | path_a_only_denominator<br>(n=72) | path_b_only_denominator<br>(n=72) | interpretation           |
|-----------------------------------------|----------------------------|-----------------------------------|-----------------------------------|--------------------------|
| <b>all_hit_at_50</b><br>CQ – Reflection | <b>-1.0</b><br>loss        | <b>-1.0</b><br>loss               | <b>-1.0</b><br>loss               | stable completeness loss |
| <b>any_hit_at_50</b><br>CQ – Reflection | <b>+0.0</b><br>tie         | <b>+0.0</b><br>tie                | <b>+0.0</b><br>tie                | stable exposure tie      |

Figure 5: LongMemEval X.4 sensitivity status. The CQ-vs-Reflection delta is stable across the primary and denominator-sensitivity cells: -1.0 on `all_hit_at_50` and 0.0 on `any_hit_at_50`.

#### 4.5 Pending Multi-Evidence Follow-Up

The follow-up tests that gap directly. `CQPendingMultiEvidence` changes the no-durable pending fallback to expose all eligible same-slot pending candidates. `ReflectionEagerWriteCardinalityCapped`

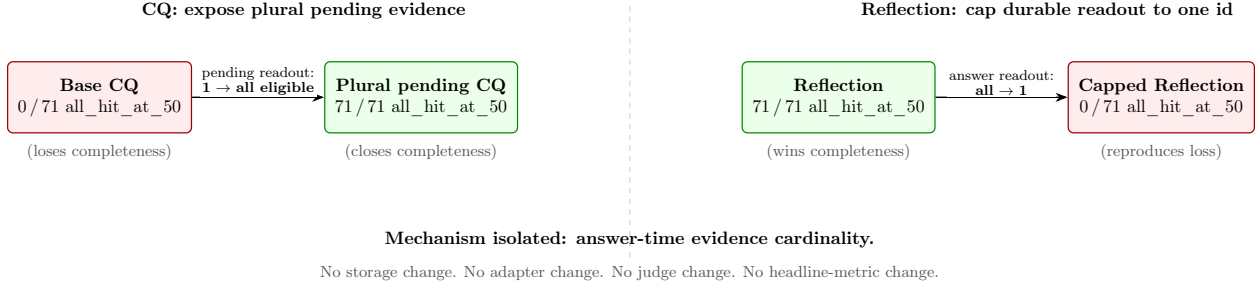


Figure 6: Phase Y cardinality control. Changing the CQ pending-readout cardinality from singleton to plural converts a 0/71 loss into a 71/71 win on `all_hit_at_50` in the primary cell. Capping the durable Reflection readout to a single id reproduces the original CQ loss with the same magnitude. The symmetric swap isolates answer-time evidence cardinality as the responsible interface variable; nothing else in the pipeline changes.

preserves Reflection’s write path and caps answer readout to one durable candidate id. Table 3 summarizes the result.

Table 3: Pending multi-evidence follow-up on `all_hit_at_50`.

| Cell                                 | Base CQ | Follow-up CQ | Reflection | Capped Reflection |
|--------------------------------------|---------|--------------|------------|-------------------|
| <code>primary_contract</code>        | 0/71    | 71/71        | 71/71      | 0/71              |
| <code>path_a_only_denominator</code> | 0/72    | 72/72        | 72/72      | 0/72              |
| <code>path_b_only_denominator</code> | 0/72    | 72/72        | 72/72      | 0/72              |

The internal regression gate reports maximum checked overall delta 0.0 across forced contradiction, preference drift, mechanism-diverse held-out, adversarial upstream noise, and evidence-conflict spectrum. Focused negative-control tests confirm that below-floor poisoned siblings, contested siblings, wrong-scope siblings, mirrored-source weak siblings, and demoted siblings are not returned by plural pending lookup. The supported mechanism is readout cardinality rather than a richer CQ-only memory substrate.

## 5 Discussion

The result is a contingency map. When the noisy stream preserves contradiction edges and the policy-query surface stays aligned, CQ’s contestation and demotion mechanism remains visible. If the stream or interface omits canonical slots, source identity, scope handles, or the relevant policy-facing identifier, policy ties become evidence about the benchmark surface rather than about policy equivalence.

This is why the negative and mixed rows are central rather than embarrassing. The 7B gate shows that aggregate component metrics can hide concentrated failures. The 32B Bucket B audit leaves several ties as bounded nulls rather than convergence evidence. The QR-canon audit shows that clustering-quality metrics can pass perfectly while exact lookup handles fail. The LongMemEval transfer shows that an external adapter can pass methodology gates and still produce a stable CQ-negative result. The pending multi-evidence follow-up then shows how a preregistered interface repair can isolate one failure mechanism without rewriting the original result.

The strongest claim is therefore methodological. Same-stream memory-policy evaluation makes

policy distinctions auditable. It can show positive mechanism survival, such as forced contradiction under 32B extraction. It can also show when a policy distinction is hidden by the interface, such as exact policy-query canonical-id failure under perfect clustering or evidence completeness failure under singleton pending readout. A useful benchmark must make both outcomes visible.

The CQ case study also illustrates why immediate-write baselines must be strong and fair. Reflection wins some rows for mechanistically uninteresting reasons, such as overwriting into a durable state that happens to include both evidence sessions. That is still a valid comparison result. The proper response is to name the mechanism, preregister a repair if warranted, and add a control that can reproduce the failure by changing only the suspected interface.

## 6 Local Unlock Probe

The local unlock probe bridges failure-taxonomy evidence and noisy policy execution. It reruns the locked 7B anchor and two 32B primary cells under preregistered model tags, schema profiles, digest checks, and frozen sentinel requirements. The 7B anchor reproduces the locked failure counts. Both 32B cells clear the preregistered gate, with zero primary scenario errors, zero primary observed failures, zero aggregate failures, and zero frozen-sentinel observed failures.

This section is intentionally short because the unlock is not itself a memory policy result. Its role is licensing. It shows that the later Phase 4 noisy policy comparison was not run on an extractor stream that the preregistration would have rejected, while the 7B anchor remains a clean example of an aggregate-passing false unlock that the full gate prevented.

## 7 Limitations

The noisy-mode support is mechanism-local. CQ separates from `ReflectionEagerWriteLite` on forced contradiction and partially on preference drift under the current 32B artifacts, ties `Mem0Lite` on the frozen sentinel, and leaves source-independence, scope-contamination, useful-pending, and poisoning-defense rows unsupported under the current evidence.

The LongMemEval result is also bounded. The completed X.4 transfer is stable and negative for base CQ on evidence completeness. The pending multi-evidence follow-up supports a readout-cardinality repair for that loss. LongMemEval-V2, LongMemEval-S, distractor-heavy retrieval quality, and natural-language QA success remain future evaluation targets; the `answer_correct` surface is diagnostic-only in X.4 because policy answers are trace-shaped.

PFLC is a diagnostic class rather than a new pass/fail gate. The field-level survey records what released baseline artifacts expose. LoCoMo’s evidence field is a clean gold-side target, and MemoryAgentBench’s conflict-resolution tasks are PFLC-relevant. The B-3 outcome is about released artifact contracts and standard output surfaces.

The CQR replay repair remains Bucket D. The QR-canon audit leaves the aborted audit and the Phase 4 Bucket B verdict unchanged, and the blocked cross-tab of QR-canon hit versus per-policy answer success remains unrun. A third CQR attempt would require either materializing the original locked run JSON payloads or preregistering a fresh Phase 4 baseline with archived payloads.

Finally, all experiments are local-first prototypes. Real-user weeks-long helpfulness, learned semantic scope inference, production retrieval stacks, and API-hosted extractors remain future evaluation targets.

## 8 Future Work

The nearest follow-up is packaging, not another rescue experiment: finish the figures, tables, claim cross-checks, reproducibility bundle, and external review pass for an arXiv technical report. The draft should keep every claim connected to an artifact path and keep unsupported claims visible in the main limitations section.

Scientifically, the next licensed experiments should be preregistered before execution. One direction is a policy-facing adapter-contract audit or a fresh Phase 4 baseline with committed or archived run JSON payloads, so the blocked CQR cross-tab can be tested without relying on missing artifacts. Another is a new external-transfer boundary, such as a MemoryAgentBench-style conflict-resolution adapter, only if the same-stream and same-substrate invariants can be preserved or the result is explicitly labeled descriptive-only.

Further LongMemEval work should treat the observed X.4 metric as fixed evidence. Any extension to LongMemEval-S, LongMemEval-V2, distractor-heavy retrieval, or additional policy interfaces needs a fresh preregistration, fixed denominator, PFLC target, judge gate, and abort criteria.

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## A Failure Taxonomy

The Phase 3 taxonomy groups component failures into four recurring modes: required-field omission, ID-namespace confusion, durable-claim drift into temporary/session framing, and contradiction-edge misses. These categories are used to explain why the 7B aggregate-passing run stayed locked and why some 32B rows remain unsafe to interpret as policy convergence.

Representative inspection traces are:

- `data/results/audit_trace_forced_contradiction_default.html`
- `data/results/audit_trace_scope_contamination_default.html`
- `data/results/audit_trace_useful_pending_memory_default.html`

The traces are qualitative inspection aids. Numeric claims come from metrics, manifests, component summaries, and compact audit evidence.

## B Reproducibility

The repository records provenance at several layers: preregistration lock hashes, adapter pins, model tags and digests, prompt/schema profiles, candidate-stream hashes, clean-worktree checks, Python version, commands, and artifact hashes. Small headline verification artifacts are committed. Large per-scenario run payloads are regeneratable-only unless a preregistration requires byte-stable replay.

The draft was prepared on branch `paper/ship-paper-draft`, based on `longmemeval-externalization-gate`. The Phase Z plan allowed either merging the LongMemEval follow-up branch into `main` or pinning the paper package to the exact branch and commit. Because `main` and `longmemeval-externalization-gate` diverged after the earlier merge, this draft uses the branch-pinning path. The reproducibility README in `paper/repro/README.md` records that decision and lists the commands for local verification.

The CQR replay aborts are reproducibility evidence. The path-normalized repair preserved stable checks and still refused to emit a verdict when locked run JSON payloads could not be reproduced. The paper treats that as support for strict replay discipline.

## C Preregistration Locks

The main preregistration and lock documents are:

- `docs/preregistered_memory_governance_evaluation_template.md`
- `docs/preregistration.md`
- `docs/local_unlock_probe_preregistration.md`
- `docs/noisy_policy_comparison_preregistration.md`
- `docs/adversarial_upstream_noise_preregistration.md`
- `docs/adversarial_upstream_noise_dated_followup_preregistration.md`
- `docs/abstention_quality_preregistration.md`

- docs/canonical\_id\_resolution\_audit\_preregistration.md
- docs/canonical\_id\_resolution\_audit\_repair\_preregistration.md
- docs/qr\_canon\_audit\_registration.md
- docs/policy\_facing\_lookup\_contract\_registration.md
- docs/fair\_stream\_externalization\_preregistration.md
- docs/cq\_pending\_multi\_evidence\_preregistration.md

The pending multi-evidence follow-up lock SHA is 84b3c44148494c1b6f72e8201a8da886f8e1f48e49ce00197f53. The canonical-id PFLC alias function remains the byte-locked CQR alias function; the PFLC work-stream does not generalize that alias rule to retrieval-id, slot-id, fact-id, or answer-handle instances.

## D Artifact Index

The most important artifact paths for this draft are:

- docs/predictions\_vs\_results.md
- docs/adversarial\_upstream\_noise\_results.md
- docs/adversarial\_upstream\_noise\_dated\_followup\_results.md
- docs/abstention\_quality\_results.md
- data/results/component\_gate\_decision\_general\_v1\_summary.json
- data/results/component\_gate\_decision\_qwen2\_5\_7b-instruct-q4\_K\_M\_default\_summary.json
- data/results/component\_gate\_decision\_qwen2\_5\_32b-instruct-q4\_K\_M\_default\_summary.json
- data/results/component\_gate\_decision\_qwen2\_5\_32b-instruct-q4\_K\_M\_scenario\_conditioned\_summary.json
- docs/noisy\_policy\_comparison\_results.md
- data/results/noisy\_policy\_comparison\_summary.json
- docs/noisy\_policy\_mechanism\_audit.md
- data/results/noisy\_policy\_mechanism\_audit\_evidence.json
- docs/canonical\_id\_resolution\_audit\_results.md
- data/results/canonical\_id\_resolution\_audit\_stop\_2026-05-15.json
- data/results/canonical\_id\_resolution\_audit\_stop\_2026-05-16.json
- docs/qr\_canon\_audit\_results.md

- data/results/qr\_canon\_audit\_metrics.csv
- docs/policy\_facing\_lookup\_contract\_proposition.md
- docs/qr\_canon\_field\_diagnostic\_results.md
- data/results/qr\_canon\_field\_diagnostic\_metrics.csv
- docs/locomo\_baseline\_replay\_survey.md
- data/results/locomo\_amb\_pflc\_summary.json
- docs/longmemeval\_transfer\_results.md
- data/external/longmemeval/transfer\_summary.json
- data/external/longmemeval/transfer\_per\_case\_rows.csv
- data/external/longmemeval/transfer\_manifest.json
- docs/cq\_pending\_multi\_evidence\_results.md
- data/external/longmemeval/transfer\_followup\_summary.json
- data/external/longmemeval/transfer\_followup\_per\_case\_rows.csv
- data/external/longmemeval/transfer\_followup\_manifest.json

Figures ship in two forms. The four conceptual diagrams (evidence ledger, externalization protocol, PFLC formal gap, phase Y cardinality control) are TikZ source files under `paper/figures/` that PDFLaTeX compiles inline at build time. The two quantitative charts (Phase 4 mechanism audit, LongMemEval X.4 cells) are matplotlib output committed as paired PNG/PDF and regenerated from the committed artifacts above by `paper/repro/render_charts.py`. All figure values trace to the documents and result JSONs listed in this index.